

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch: Mechanical Engineering Semester : Subject Code & Name: BTMEC506A Automobile Engineering Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Give detailed classification of automobile.	(L1/CO1)	6
B)	Draw transmission layout of an automobile.	(L2/CO2)	6
C)	Draw and explain tractive efforts and performance curves.	(L2/CO2)	6
Q.2	Solve Any Two of the following.		
A)	List different types of independent suspension systems and explain any one with neat sketch.	(L2/CO2)	6
B)	Draw and explain layout of steering system.	(L2/CO2)	6
C)	With neat sketch explain Davis steering mechanism.	(L2/CO2)	6
Q. 3	Solve Any One of the following.		
A)	With neat sketch explain the working of constant mesh gear box.	(L2/CO2)	6
B)	What is differential. Explain its working principle.	(L2/CO2)	6
C)	Write a note on fluid coupling and torque convertors.	(L2/CO2)	6
Q.4	Solve Any Two of the following.		
A)	Explain the working of disk brake and drum brake with neat sketch.	(L2/CO2)	6
B)	Differentiate between radial and cross-ply tyres.	(L2/CO2)	6
C)	List and explain different types of wheels.	(L2/CO2)	6
Q. 5	Solve Any One of the following.		
A)	With neat sketch explain the construction and working of ignition system.	(L2/CO2)	6
B)	Explain the construction and working principle of lead acid battery.	(L2/CO2)	6
C)	Write a note on emission control system and regulations.	(L2/CO2)	6
	*** End ***		

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